

Flood Polynomial Model

MKM-FPO-001 — A Closed-Form Surrogate for the Gauge Flood Classifiers

MKM Research Labs

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1 Executive Summary

The Flood Polynomial Model (MKM-FPO-001) gives the probability that a gauge goes on to flood, from its water level and the hour of the storm, in closed form. It stands in for the per-gauge gradient-boosted classifiers (MKM-FC-001) where a stress scenario needs $P(\text{flood})$ without loading a trained model for every gauge.

This is version 2.0, a new model. Version 1.0 was benchmarked against an independent Wolfram calculation and against the classifiers it approximates, and it failed. At hour 24 it fell as the water rose between 52% and 74% of the severe level. At the severe level it gave 0.00 at hour 0 and 0.29 at hour 167, where its document printed 0.42 and 0.97. Eight of the 54 cells of its printed table were reproduced by the code, and none of its accuracy figures held. It had been fitted to one gauge.

Version 2.0 changes the model in four ways:

- **At or above severe, $P = 1$.** Reaching the severe level is the flood the classifiers are trained to predict. Below it, P is a fitted logistic.
- **Fitted to every gauge.** One kernel, cubic in the two features, fitted to all 11 gauges' classifiers on 1,807,233 real storm states and measured on 1,807,896 held-out ones.
- **Rising water never lowers P.** The fit is constrained to make P non-decreasing in the level at every hour.
- **No extrapolation early in a storm.** No storm reaches 40% of severe before hour 4.7, or severe before hour 37.4. Earlier than that the classifiers have never seen a state, so the model holds the hour at that envelope instead of extrapolating.

On held-out storms near severe ($w/s \geq 0.75$), the mean absolute difference from each gauge's classifier is 0.0408 to 0.0658 across the 11 gauges, and the bias is -0.034 to 0.025. The difference exceeds 0.1 in 10% to 21% of those states. That is the price of a formula that sees only the level and the hour: the classifiers also see how fast the water is rising, and no function of level and hour can follow that.

2 Model Purpose and Scope

2.1 Purpose

The model answers one question in closed form: given a gauge at water level w in hour n of a storm, how likely is it to reach its severe threshold s? The stress scenarios ask it hour by hour along a hydrograph, for many gauges, where loading a trained classifier per gauge is not practical.

2.2 Scope

- **In scope:** $P(\text{flood})$ from w/s and the storm hour, for any gauge with a severe threshold; its slope in the level, for sensitivity.
- **Out of scope:** the rate of rise and other features the classifiers use; gauge-specific behaviour; spatial correlation between gauges.

2.3 How Production Uses It

- **Single-gauge stress scenario** (`src/routes/trading/stress/scenario.py`) evaluates `p_flood_poly` at each hour of the scenario hydrograph, and prints a probability surface over levels and hours. Once the level has reached severe, the scenario holds $P = 1$ for the rest of the storm, even if the water falls back: the flood has happened, and the swaps it triggers are knocked out. That rule is the scenario's, not the model's.
- **Portfolio stress screen** (`src/static/js/trading/port_stress/setup.js`) draws $P(\text{flood})$ hour by hour in the browser. It fetches the model from `GET /api/v1/trading/stress/flood-poly` and evaluates it there. Until v2.0 it carried its own copy of v1.0's coefficients.
- **Portfolio stress P&L** (`src/routes/trading/port_stress/_compute.py`) uses `p_flood_simple`, which is $\min(1, w/s)$. That is a proximity ratio, not a probability (§6.3).
- **Sensitivity** (`src/models/stress/sensitivity.py`) evaluates the model and its slope over candidate errors in the severe threshold.

2.4 Model Chain Position

MKM-FC-001 (per-gauge classifiers) $\xrightarrow{\text{fitted to}}$ **MKM-FPO-001** $\rightarrow$ stress scenarios (MKM-ST-001)

3 Mathematical Framework

3.1 Features

Two gauge-independent features, as the classifiers use them:

$$h = \ln\left(\max\left(\frac{w}{s},\ \phi\right)\right), \qquad \phi = 0.2, \tag{1}$$

$$t = \ln\left(\frac{n+1}{T}\right), \qquad T = 168, \quad n \in \{0, \ldots, 167\}. \tag{2}$$

h is zero at the severe level and negative below it. t runs from $\ln(1/168) \approx -5.12$ at hour 0 to zero at hour 167. Below the level floor ϕ, 20% of severe, the level is held at the floor.

3.2 Model Equation

$$P(\text{flood}) = \begin{cases} 1 & w \geq s, \\ \sigma\big(z(h,\ \max(t,\ \tau(h)))\big) & w < s, \end{cases} \qquad z(h,t) = \sum_{k=1}^{10} c_k\, h^{i_k} t^{j_k}, \tag{3}$$

where $\sigma(z) = 1/(1+e^{-z})$ with z clamped to $[-30, 30]$, the ten terms (i_k, j_k) are every power of h and t up to total degree three, and $\tau(h)$ is the support envelope of §3.3.

At or above severe. A gauge at its severe level has flooded: that is the event the classifiers are trained to predict. So $P = 1$ there, at every hour. Version 1.0 had no such rule and fell back to 0.29 at the severe level late in a storm; the stress route had been setting $P = 1$ there itself, without the document saying so.

Below severe. The cubic kernel is fitted to the classifiers (§4.1). From hour 16 on, at a fixed level, P never rises as the storm goes on: a level reached late in a storm is one the water has been slow to reach, and it is less likely to go on to severe. That is what the classifiers say, and the reverse of what v1.0's document described. In the first 16 hours, just after the envelope, P can rise with the hour before it falls, by at most 0.06.

3.3 The Support Envelope

The kernel is fitted where storms go. Storms rise at a finite rate, so a level is first reached some hours into a storm, and the classifiers have never been shown a state earlier than that. There, the polynomial is extrapolating, and a cubic extrapolates freely: refitted without a guard, it gave 0.998 at half of severe at hour 0, where no storm has ever been.

$\tau(h)$ is the earliest log time at which any gauge's storms reached the level h, measured over 2,000 storms at each of the 11 gauges (Table 1), and interpolated linearly in t between knots. Below the first knot, 32.6% of severe, every hour is supported. Earlier than the envelope, the model holds the hour there: a level reached sooner than any storm reached it is priced as the earliest storm that did. The cells this affects are marked in Table 4.

w/s	earliest hour	w/s	earliest hour
0.326	0.0	0.7	15.1
0.35	2.3	0.75	18.4
0.4	4.7	0.8	22.0
0.45	6.1	0.85	24.9
0.5	7.5	0.9	29.3
0.55	8.9	0.95	35.7
0.6	10.8	1	37.4
0.65	12.9		

Table 1: The support envelope: the earliest storm hour at which any gauge's storms reached each fraction of severe. Earlier than this the hour is held here; between knots, linearly in t.

3.4 Monotonicity

Rising water never lowers P, at any hour. Three parts make that hold:

- Between the floor and severe, the fit holds $\partial z/\partial h \geq 0.01$ on a grid of 120 levels and all 168 hours.
- Where the hour is held at the envelope, P moves with the level along the envelope, at rate $\partial z/\partial h + \partial z/\partial t \cdot \tau'(h)$. The fit holds that at least 0.01 too, at 20 points on each segment.
- Below the floor P is constant, and at severe it steps up to 1.

The smallest constrained slope in the fit is 0.0100. The test suite checks P is non-decreasing over 2,000 levels at every one of the 168 hours.

The hour is a whole number (Eq. (2)), and every caller passes one. That matters in one place. Between hour 0 and hour 1, where t moves fastest, the kernel's slope in the level dips below zero near a third of severe (to about -0.10 at hour 0.4). No whole hour falls there, so the model never evaluates it, but the kernel is not monotone at fractional hours and should not be called with one.

4 Input Parameters and Calibration

4.1 Calibration Data

The target is each gauge's production classifier, read as production reads it, on the real storm states it is trained on: 2,000 storms per gauge, drawn once with seed 7 by the challenger's catalogue loader, for all 11 gauges with a classifier. Only states below severe are used, because at and above it $P = 1$ by definition. Half of each gauge's storms are fitted and the other half measured. That gives 1,807,233 fitted and 1,807,896 held-out states.

4.2 Fitting

`python -m tools.flood_poly_fit` minimises the weighted squared difference between $\sigma(z)$ and the classifier's probability, pooled over every gauge, with the monotonicity penalty above. States at or above 75% of severe weigh five times as much, because that is where a stress scenario's outcome is decided. The fitted states are evaluated as the model evaluates them, with the level floor and the envelope applied.

The optimiser's tolerances matter. At SciPy's defaults (10^{-8}) the fit stopped on a plateau after 94 evaluations, and on synthetic data it failed to recover a kernel it had been shown. At 10^{-15} it recovers it exactly. The tolerances are in `config.models` and the test suite checks the recovery.

4.3 Coefficients

k	Term $h^i t^j$	c_k
1	h	+32.871867
2	t	−12.801048
3	ht	−7.540193
4	h^2	+44.826451
5	t^2	−2.848226
6	h^3	+54.512653
7	t^3	−0.756589
8	h^2t	−29.568283
9	ht^2	+6.254275
10	1	−1.293810

Table 2: Fitted coefficients, as `config.models` holds them.

4.4 Model Parameters

Parameter	Config name	Value
Level floor ϕ	`flood_poly_level_floor`	0.2
Storm horizon T	`flood_poly_storm_hours`	168
Exponent clamp	`flood_poly_exp_clamp`	30
Kernel terms	`flood_poly_terms`	10
Support envelope	`flood_poly_envelope`	15 knots

5 Fit Quality

Measured on the held-out storms, against each gauge's own classifier:

Gauge	Near MAE	Near bias	Near > 0.1	Below MAE
GAUGE-085522ee	0.0590	+0.0247	20%	0.0125
GAUGE-1aaf8b08	0.0408	−0.0070	10%	0.0087
GAUGE-9e5f3ef9	0.0542	+0.0128	16%	0.0198
GAUGE-9fffb377	0.0658	−0.0343	21%	0.0203
GAUGE-a6be4ee9	0.0465	+0.0171	15%	0.0075
GAUGE-b2dbe58b	0.0518	+0.0087	17%	0.0100
GAUGE-c0fbc443	0.0478	−0.0015	15%	0.0064
GAUGE-d208bf3d	0.0557	−0.0081	16%	0.0179
GAUGE-dc5e88fb	0.0578	+0.0246	21%	0.0092
GAUGE-ef53398c	0.0628	−0.0288	20%	0.0166
GAUGE-f8629444	0.0566	+0.0188	18%	0.0174

Table 3: Agreement with each gauge's GBM on the held-out storms (1,807,896 states). "Near" is $w/s \geq 0.75$; "> 0.1" is the share of near states where $|P - \text{GBM}|$ exceeds 0.1.

Below severe overall the mean absolute difference is 0.0064 to 0.0203, because most states are far below severe and both models give nearly zero there. Near severe the difference is 0.0408 to 0.0658. The bias near severe is −0.034 to 0.025: the model is not conservative everywhere, and it under-predicts at some gauges.

The difference that remains is mostly the rate of rise. The classifiers give different probabilities to two states at the same level and hour, one rising fast and one barely rising. A function of level and hour can at best give their average.

6 Implementation Details

6.1 Source Code

- `src/models/stress/flood_poly.py`: `p_flood_poly`, `p_flood_series`, `p_flood_simple`, the features and envelope, `model_card`, `client_params`.
- `config/models/_misc.py`: coefficients, terms, floor, envelope and fit settings.
- `src/routes/trading/stress/flood_poly.py`: serves `client_params` to the browser.
- `src/static/js/trading/port_stress/setup.js`: the browser's evaluation of the served model.
- `tools/flood_poly_fit/`: the fit.

6.2 Usage

```
from models.stress.flood_poly import p_flood_poly, p_flood_series

p = p_flood_poly(water_level=5.2, hour=48, severe_level=6.58)
ps = p_flood_series([4.0, 4.5, 5.2, 5.8, 5.5], severe_level=6.58)  # hour = position
```

6.3 `p_flood_simple` Is Not a Probability

`p_flood_simple(w, s)` returns $\min(1, w/s)$. It is a proximity measure: a gauge at half its severe level scores 0.5, where Table 4 gives a probability anywhere from nearly zero to 0.66 depending on the hour. Portfolio stress P&L uses it where only a gauge's peak level is known. Its name and v1.0's document presented it as a time-independent $P(\text{flood})$; it is kept, under that name, as a ratio, and is documented as one.

6.4 Numerical Safeguards

- The level floor keeps $\ln(w/s)$ finite at zero or negative water levels.
- The exponent is clamped to $[-30, 30]$ before the logistic.
- A severe threshold at or below zero gives $P = 0$.

6.5 Dependencies

The model uses only the Python standard library at run time.

7 Sensitivity Analysis

7.1 Probability Surface

w/s	hr 0	hr 6	hr 12	hr 24	hr 48	hr 84	hr 120	hr 167
0.20	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.30	0.046	0.029	0.001	0.000	0.000	0.000	0.000	0.000
0.40	0.334†	0.388	0.352	0.043	0.000	0.000	0.000	0.000
0.50	0.608†	0.608†	0.664	0.490	0.034	0.000	0.000	0.000
0.60	0.896†	0.896†	0.891	0.801	0.293	0.008	0.000	0.000
0.70	0.990†	0.990†	0.990†	0.969	0.733	0.079	0.003	0.000
0.80	0.999†	0.999†	0.999†	0.999	0.970	0.461	0.036	0.001
0.90	1.000†	1.000†	1.000†	1.000†	0.999	0.937	0.348	0.013
0.95	1.000†	1.000†	1.000†	1.000†	1.000	0.988	0.714	0.054
0.99	1.000†	1.000†	1.000†	1.000†	1.000	0.997	0.906	0.165
1.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Table 4: $P(\text{flood})$ by level and storm hour, from `p_flood_poly` ($s = 6.58$ m; the model depends on w/s only). At $w/s = 1$ a flood has happened. Below 0.20 the level is held at 0.20. † Earlier than any storm reached the level: the hour is held at the envelope (Table 1).

Three things to read from Table 4. At a fixed level, P falls as the storm goes on, after its first hours. Every row rises with the level. And at the severe level P steps to 1: just below it, late in a storm, P is well below 1, so an error in the severe threshold that moves s across the water level moves P by the whole of the remaining gap.

7.2 Severe-Threshold Error

The threshold s is per-gauge and enters only through h, so a relative error e in s shifts h by about $-e$, and moves P by about $-e\,\partial P/\partial h$. `stress.sensitivity.threshold_sensitivity` reports this, using the model's own closed-form slope, including along the envelope where the hour is held.

8 Validation and Backtesting

8.1 What the v1.0 Benchmark Found

The Wolfram benchmark of v1.0 (MKM-FPO-001_wolfram_benchmark v0.1, 03-Nov-2026) found the code computed its document's equations exactly (worst relative error 1.8×10^{-14}), but the document's claims did not hold:

- P fell as the water rose: at hour 24 it passed $\frac{1}{2}$ at both 52% and 74% of severe.
- 8 of Table 1's 54 cells and none of Table 2's six were reproduced.
- At the severe level P was 0.00 at hour 0 and 0.29 at hour 167, not 0.42 and 0.97.
- The accuracy claim $|P - \text{GBM}| < 0.05$ failed: near severe the 95th percentile reached 0.26.
- The stress route set $P = 1$ at severe without the document saying so.
- `p_flood_simple` is a ratio, not a probability.

Each of those is answered above: the step at severe in Eq. (3), the constrained fit, the regenerated tables, the measured accuracy, and §6.3.

8.2 Implementation Benchmark

Version 2.0 is benchmarked in the same way (MKM-FPO-001_wolfram_benchmark v0.2): the code against an independent 50-digit reference written from this document, Wolfram against the reference, and this document's tables and claims against the code and the classifiers.

9 Model Limitations and Known Weaknesses

9.1 Assumptions

1. **FPO-A1: One kernel for all gauges.** The features normalise each gauge by its own severe threshold, and one kernel is fitted to all 11 gauges together. Accuracy differs by gauge (Table 3).
2. **FPO-A2: Level and hour suffice.** The rate of rise is left out. It is the main source of the remaining difference from the classifiers.
3. **FPO-A3: The envelope is the support.** The envelope is measured on the storm catalogue. A real storm that rises faster than any in the catalogue is priced as the fastest one that did.

9.2 Limitations

1. **FPO-L1: Accuracy near severe.** The difference from the classifiers exceeds 0.1 in 10% to 21% of states near severe.
2. **FPO-L2: Not conservative everywhere.** The bias near severe is negative at some gauges.
3. **FPO-L3: The step at severe.** P jumps to 1 at the severe level, so results near it are sensitive to the threshold (§6.3 and the sensitivity section).
4. **FPO-L4: Inherits the classifiers.** It approximates MKM-FC-001 and carries whatever error the classifiers carry.

10 Governance Validation Questions

The following nine questions are mandated by Chapter 5 of the *Model Risk Governance for Vendors* handbook and must be explicitly addressed for every model in the inventory. Response status is tracked in the Model Governance Dashboard (MRC portal).

10.1 Q1 — Ronseal Test

Yes, now. v1.0 did not do what its document said; v2.0 is documented from the code and its tables are generated from it.

10.2 Q2 — Accuracy and Stability

Near-severe mean absolute difference 0.0408–0.0658 against the classifiers on held-out storms. The surface is monotone in the level by construction.

10.3 Q3 — Resource Usage

Negligible: ten coefficients and the standard library.

10.4 Q4 — Override Controls

None in the model. Coefficients and envelope are configuration, changed by refitting.

10.5 Q5 — Output Consumption

Stress scenarios, the portfolio stress screen and the sensitivity diagnostic.

10.6 Q6 — Model Chaining

Yes: it approximates MKM-FC-001 and feeds the stress pipeline (MKM-ST-001).

10.7 Q7 — Weaknesses

Accuracy near severe, the omitted rate of rise, and the step at severe.

10.8 Q8 — Monitoring

Refit and re-measure when the classifiers are retrained or the storm catalogue changes; re-run the implementation benchmark on any change to the model or its configuration.

10.9 Q9 — Overall Risk

Acceptable for stress testing, where it gives a consistent monotone surface across gauges. It should not stand in for a gauge's classifier where that gauge's own probability is needed.

11 Change History

- **v2.0:** a new model after the v1.0 benchmark: $P = 1$ at or above severe; a cubic kernel fitted to all 11 gauges' classifiers on real storm states, constrained monotone in the level; the level floor; the support envelope; the browser served the model instead of a copy; `p_flood_simple` documented as a ratio; tables generated from the code.

12 Document History

Date	Version	Description	Author
28-Mar-2026	1.0	Initial documentation: quadratic kernel fitted to GAUGE-dc5e88fb	D. Kelly
03-Nov-2026	2.0	New model after the Wolfram benchmark of v1.0; re-fitted to every gauge	David K Kelly